Rai Yashmit

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SKILLS

Languages: C++, C, Python, SQL
Tech Stack: Power BI, MySQL Workbench, Excel, Docker, Git
Frameworks: TensorFlow, Scikit-Learn, Fast API, NumPy, Pandas
Core Skills: Data Structures and Algorithm, System Design, Machine Learning, Data Analysis
Soft Skills: Strategic Thinking, Adaptability, Rapid Skill Acquisition, Leadership

PROJECTS

Denoising Diffusion Probabilistic Model (DDPM) | Python, TensorFlow, UNet — Dec' 25

- Engineered a conditional DDPM with 1,000 diffusion timesteps, trained on 60K MNIST images using TensorFlow.
- Reduced reconstruction loss by **~10%** and maintained a train–validation loss gap **<0.001,** indicating strong **generalization**
- Achieved **stable convergence with early stopping**, reaching a best **validation MSE of ~0.039.**

Sorting Visualizer | Python, FastAPI, Docker, Collections — Jun' 25

- Engineeried a backend computation engine to execute and visualize **12+ algorithms (sorting and graph-based)** with step-by-step state outputs.
- Automated delivery of educational metadata for each algorithm, serving **30+ unique data points per request.**
- Implemented optimized data structures and algorithms including Quick Sort, Merge Sort, Heap Sort, Dijkstra's, BFS, DFS, Prism, Kruskal and etc.
- Containerized the backend application using **Docker and docker-compose**, ensuring consistent development and deployment across local and production environments

TRAINING

Library Management System | Python, SQLite, FastAPI — Aug' 25

- Architected a RESTful backend system using FastAPI, handling 20+ endpoints for user, inventory, and transaction management.
- Structured the application using **Repository and Service Layer patterns** to decouple business logic from database operations, enabling scalable and modular system design.
- Designed a normalized relational schema (3NF) supporting **5+ entities and complex relationships (1:N, M:N)**.
- **Integrated a user preference engine** supporting multi-attribute subscriptions and implemented an **Observer Pattern-based notification system** for real-time book availability alerts

CERTIFICATES AND CERTIFICATIONS

- Data Science | Oracle — Oct' 25
- Data Structures and Algorithm | LPU — Jul' 25
- Elite in Cloud Computing | NPTEL — May' 25

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelors of Technology in Computer Science and Engineering — Aug' 23 - Present
CGPA – **8.84**